

Arvin Munkh-Ochir

arvin.munkh-ochir@stonybrook.edu | 12arvin09.io/Arvin/ | github.com/12arvin09

Skills

Programming Languages : JAVA, Embedded Rust, React.js, LuatOS, HTML, CSS, JavaScript, Python, C++, C

Engineering Skills: Hardware : Transmission Protocols: UART, SPI, I²C

Software & Tools: OpenCV, NumPy, OpenBCI, OpenVibe, CorelDRAW, InkScape

Experience

Machine Learning Intern, AI Academy Asia

Jun 2026 – Aug 2026

- Developed and fine-tuned a Mongolian speech-to-text model using Whisper, Hugging Face, LoRA, and QLoRA on 50+ hours of collected, transcribed, and cleaned audio data.
- Improved WER on the Google FLEURS benchmark from 20.2% to 17.4% (13.9% relative reduction) through model benchmarking, hyperparameter tuning, and GPU-based training experiments.

Selected Participant, Annual Nepal AI School, NAAMII – Kathmandu, Nepal

Dec 2025 – Jan 2026

- Selected as one of 210 participants from 745 applicants across 35 countries for an intensive international AI program.
- Participated in advanced lectures and lab sessions on geometric deep learning, graph neural networks, transformers, reinforcement learning, computer vision, and large language models

Software Engineering Intern, OndoSpace LLC – Ulaanbaatar, Mongolia

Jul 2025 – Aug 2025

- Researched AI applications at the intersection of space and telecom at ONDOSpace, initiating the ongoing GerGrid AI project.
- Launched SugarBox, an internal feedback platform built with React and Next.js, adopted by 80% of ONDO LLC employees.

Electrical and Computer Engineering Intern, OndoSpace LLC – Ulaanbaatar, Mongolia

Jun 2024 – Dec 2024

- Improved motion sensing accuracy for 0.5 U Cube Satellite Development to receive accurate calibrated data from 9-axis IMU Sensor, by programing them on Embedded Rust.
- Evaluated the data usability for Satellite development executing test comparison codes on BMX160, HW038, MCP9808 STM32 processors and sensors, by coding them on Embedded Rust.

Projects

GerGrid AI Infrastructure Planning Start-Up

- Explored the usage of AI based models to optimize cell tower placement in Mongolia's mountainous terrain by integrating existing LLM models with telecommunication platforms like Atol. Included ONDOSpace's LEO orbited cube satellites feature.

Spelling Letters Through Brain Waves

[Arvin/projectbci](#)

- Explored brain computers interface with non-invasive EEG technology to detect letters from brain signals. Implementing the P300 speller paradigm of OpenVibe Software, achieved accuracy improvement from 72% to 82% through repeated training.

Location Detecting Smart Backpack

[Arvin/projectcane](#)

- Developed a smart backpack for visually impaired users with integrated location sensing, time announcements, and location sharing through programming sensors with LuatOS.

Smart Cane for Visually Impaired People

[Arvin/projectbackpack](#)

- Built an early smart cane prototype for visually impaired users, integrated five infrared sensors and a water sensor with a microcontroller. Enabled real time audio feedback to by using embedded system's async features.

Hamtdaa - Content Creation

[Arvin/Hamtdaa](#)

- Self driven initiative to improve rural areas education by creating content that combines international level education with Mongolian language.

Education

School – Stony Brook University | BS in Computer Science

June 2029

School – Orchlon International High School

June 2023